Ali Nikkhah

Machine Learning Engineer / Researcher

Tehran, Iran · Remote-friendly · Open to Relocation alinkkh9@gmail.com 0991 296 3951
alinikkhah2001.github.io alinikkhah2001 AliNikkhah2001

Summary

Applied ML researcher & engineer (Sharif B.Sc. Electrical Engineering 2020-2025, GPA 3.65; M.Sc. AI from Sep 2025) specialising in agentic RAG, multimodal vision-language, self-hosted LLM serving, and GPU runtime optimisation. Shipped LLM + CV pipelines at marketplace scale, self-hosted Graph RAG assistant for a national credit bureau, and BNPL risk platforms. Research focus: LLM reasoning and hallucination mitigation.

Education

M.Sc. Artificial Intelligence **Sharif University of Technology**, M.Sc. Artificial Intelligence — Tehran, Iran, Sept 2025 – present
- Computer Engineering Department, AI major (full-time, ongoing, no thesis yet)

B.Sc. Electrical Engineering **Sharif University of Technology**, B.Sc. Electrical Engineering — Tehran, Iran, Sept 2020 – Sept 2025
- GPA: 3.65
- Thesis: Team thesis on efficient design and implementation of spread-time CDMA ASIC (my slice: FPGA implementation and pre-chip FPGA tests), supervisor Dr. Fatemeh (Noyan) Akbar, grade 20/20
- Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Science, Embedded Systems, Signals and Systems, Microprocessor Systems

Experience

Iran Credit Scoring Bureau, ML Engineer — Self-Hosted Graph RAG Assistant — Tehran, Iran — Full-time · On-site, Apr 2026 – present, 6 months

Self-hosted Graph RAG company assistant for national credit scoring — Neo4j knowledge graph, self-hosted LLM runtime on GPU, tool-calling agent.
- Architected self-hosted Graph RAG assistant: Neo4j knowledge graph + vector retrieval fused into an agent that answers analyst queries over credit-scoring domain data with grounded citations.
- Ran self-hosted open-weight LLMs (vLLM/TGI) on GPU; profiled and optimised GPU runtime — KV-cache, batching, quantization (AWQ/GPTQ) — cutting per-token latency and GPU memory footprint.
- Implemented Neo4j tool-calls (Cypher generation + schema-aware retrieval) so the assistant executes graph traversals instead of hallucinating structured facts.
- Added hallucination guardrails — retrieval grounding, citation enforcement, deterministic fallback — for a regulated financial environment.
- Stack: Python, Neo4j, Cypher, Graph RAG, LangGraph, LangChain
- Most recent Iran position; overlaps Sharif M.Sc. AI (Sep 2025) — full-time industry alongside full-time graduate study

Turquoise Digital, Senior Data Scientist & Data Engineer

Tehran, Iran — Full-time · On-site
Sept 2025 – Mar 2026
7 months

Directed AI for regional retailers — multilingual reco, BNPL credit-risk ensemble (AUC 0.87), agentic support chatbot, centralized data lake + observability.

- Centralized lake from raw CDC via PeerDB + Trino/Parquet on 5TB/day, <60s freshness; Airflow + PySpark DAGs cut memory 60%, governed by Great Expectations + Prometheus/Grafana.
- Built default-risk ensemble (XGBoost + time-series Transformers) — AUC 0.87, −22% defaults; 50+ concurrent mSPRT A/B tests with canary & auto-rollback, presented to C-suite.
- Shipped Whisper/wav2vec2.0 + LangGraph agentic chatbot — −78% ticket volume, 94% intent accuracy; deterministic fallback + policy guardrails for auditability.
- Drove portfolio allocation & risk engine (Sharpe +0.42) with streaming ClickHouse signals; LlamaIndex retrieval graph for ranking features.
- Stack: Python, PyTorch, Spark, Trino, PeerDB, Airflow
- Concurrent with Advanced Analytics Australia (Remote Contract) — 7-month overlap, split focus: Turquoise BNPL/risk/whisper vs Advanced edge CV

Advanced Analytics Australia, Computer Vision / ML Platform Engineer (Edge)

Remote · Australia (Tehran-based) — Remote Contract
Sept 2025 – Apr 2026
8 months

Edge CV platform across 12 retail sites, 100+ cameras @30 FPS, Kafka streaming, quantized TensorRT/TF Lite + Triton serving.

- Deployed YOLO + ViT quantized (TensorRT/TF Lite) on Triton — 99.2% uptime <50ms p99, 3.5x speedup, exactly-once Kafka, 30 FPS over 100+ streams.
- Built edge MLOps with Triton model repository, zero-downtime rolling updates, automated rollback, Prometheus/Grafana per-site health.
- Cut false positives −35% via attention-guided NMS + SNR monitoring; broker architecture for 12 distributed sites.
- Stack: Python, OpenCV, YOLO, ViT, TensorRT, TFLite
- Parallel remote contract alongside Turquoise — edge CV track

Digikala, AI Engineering R&D Lead

Tehran, Iran — Full-time · On-site
Jan 2025 – Aug 2025
8 months

Owned agentic RAG platform for marketplace listing intelligence — modular control pipelines (MCP), hybrid retrieval, LLM serving at 5M+ queries/day.

- Led team of 5 defining R&D roadmap + SLOs; shipped MCP with LangGraph cyclic graphs, zero-downtime hot-swapping, health-checked agent composition with deterministic fallback.
- Hybrid retrieval FAISS + Elasticsearch RRF + HyDE + KAG — +23% recall@10, −41% hallucination; multi-tier Redis caching, p99 <200ms.
- LLM serving on vLLM (PagedAttention) + Triton — 3.2x throughput, 60% cost reduction; 200 pods, MIG/HPA/VPA, Prometheus KV-cache dashboards.
- Partnered with infra/product establishing eval pipelines, A/B guardrails, canary rollouts.
- Stack: Python, LangGraph, LangChain, LlamaIndex, FAISS, Elasticsearch
- Full-time lead; UBC/Trinity/Sharif were part-time remote (<10h/wk)

Digikala, AI Engineer

Tehran, Iran — Part-time · Hybrid
Nov 2024 – Jan 2025
3 months

Prototyped MCP + RAG scaffolding that scaled to org-wide mandate.

- Implemented LangGraph orchestration layers with health monitoring; hybrid retrieval with RRF + deterministic fallbacks for auditability.
- Embedded eval telemetry, memory-efficient indexing, response caching; load-tested to 5M queries.
- Stack: LangChain, LlamaIndex, FAISS, Elasticsearch, FastAPI
- Promotion track — converted to R&D Lead Jan 2025

HomaCloud, MLOps Engineer (Founding) — Tehran, Iran — Full-time · On-site — Aug 2021 – Feb 2022 — 7 months

Built K8s-native training, CI/CD, and serving infra from scratch — Docker/K8s, Ray, MLflow, FastAPI/TF Serving, Terraform, ArgoCD.

- Provisioned multi-node K8s via Terraform; Ray + MLflow training pipelines, TF Serving + FastAPI — <10 min releases, 99.9% deploy success, 15+ models.
- GitOps with GitHub Actions, canary, spot-instance optimization; Prometheus/Grafana + ELK observability; registry + lineage.
- Stack: Kubernetes, Docker, Ray, MLflow, Terraform, ArgoCD
- During B.Sc. — clean sequential after Mapna internship

Publications

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning — Jan 2025

Ali Nikkhah, [Co-authors TBD], A. Supervisor

(arXiv preprint — in preparation)

Teaching

Natural Language Processing (Graduate)
TA · Sharif (Aug 2023 – Jan 2024)

Generative Models (Graduate)
TA · Sharif (Aug 2023 – Jan 2024)

Deep Learning (Graduate)
TA · Sharif (Jan 2023 – Aug 2023)

Analog Electronics
TA · Sharif (Aug 2022 – Jan 2023)

Machine Learning & Data Science
TA · Sharif (Jan 2022 – Aug 2022)

Circuit Theory & Engineering Probability
TA · Sharif (Jan 2021 – Aug 2021)

Skills

Programming: Python, C++, Go, SQL, R, Bash, TypeScript

ML & Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, OpenCV, YOLO, ONNX

Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE, KAG, Graph RAG, vLLM, Triton, Neo4j, Cypher

LLM Serving & GPU: vLLM, TGI, AWQ, GPTQ, KV-cache, GPU Optimization, PagedAttention, TensorRT-LLM

Data & MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray, MLflow, W&B, ArgoCD, Terraform

Languages: Persian (Native), English (C2 IELTS 8.0), Arabic (Professional), French (Limited), German (Limited)